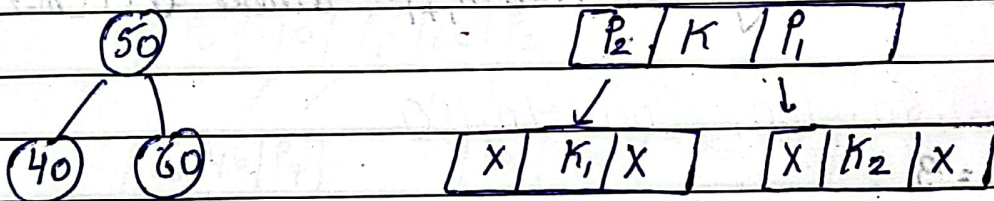


Time complexity of build a heap = $O(\log n)$

Unit - III

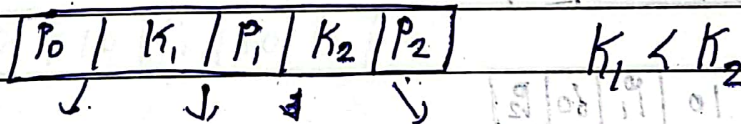
M-way Search tree

It is similar to binary search tree but it has m pointers and $m-1$ keys in one node.



Where x denote null pointer

3-ways



M-way Search tree properties

1. Each node can point at most m subtree
2. The no. of keys in each node are maximum $m-1$ and each node can have maximum m pointers.

3) M-way Search Tree representation

$|P_0| K_1 |P_1| K_2 |P_2| \dots |P_{m-1}|$

where each P_i is the pointer to subtree or null and $0 \leq i \leq m-1$

$K_1 < K_2 < \dots < K_{m-1}$ are keys

K_{j+1} are keys $0 \leq j \leq m-2$

(4) All keys values in the subtree pointed by P_i will be less than K_{i+1} and key pointed P_{i+1} will be greater than K_{i+1} where $0 \leq i \leq m-2$

Ex - 10, 60, 100, 200, 40, 120

$m=3$

I

$|P_0| 10 |P_1|$

II

$|P_0| 10 |P_1| 60 |P_2|$

III

$|P_0| 10 |P_1| 60 |P_2|$

$|P_3| 100 |P_4|$

IV

$|P_0| 10 |P_1| 60 |P_2|$

$|P_3| 100 |P_4|$

10, 60, 100, 200, 40, 120

V

$P_0 | 10 | P_1 | 60 | P_2$

$P_3 | 100 | P_4 | 200 | P_5$

VI

$P_0 | 10 | P_1 | 60 | P_2$

$P_6 | 40 | P_7$

$P_3 | 100 | P_4 | 200 | P_5$

VII

$P_0 | 10 | P_1 | 60 | P_2$

$P_6 | 40 | P_7$

$P_3 | 100 | P_4 | 200 | P_5$

$P_6 | 120 | P_7$

$X | 10 | P_1 | 60 | P_2$

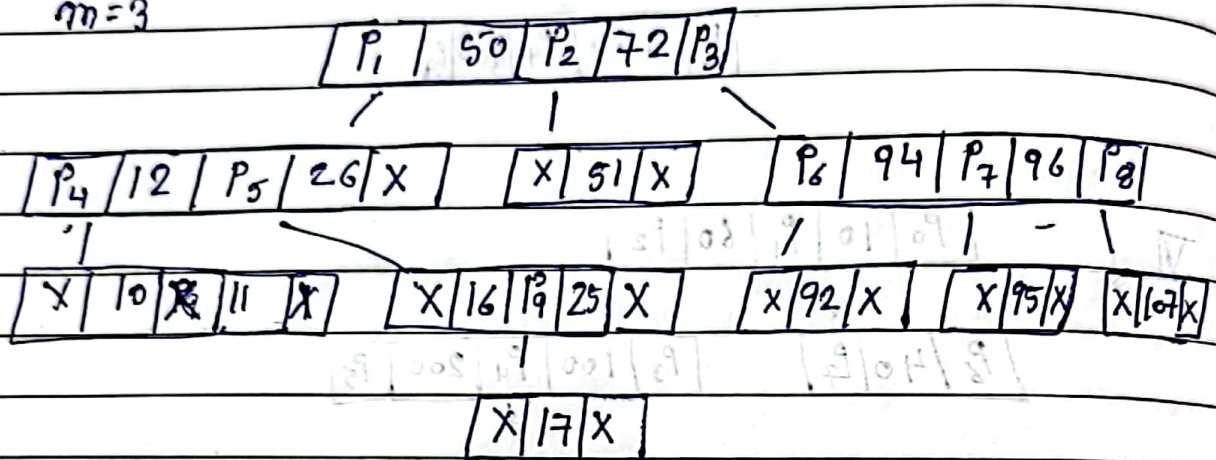
$X | 40 | X$

$X | 100 | P_4 | 200 | X$

$X | 120 | X$

Ex - 50, 72, 96, 94, 107, 26, 12, 11, 92, 10,
25, 51, 16, 17, 95

$m=3$



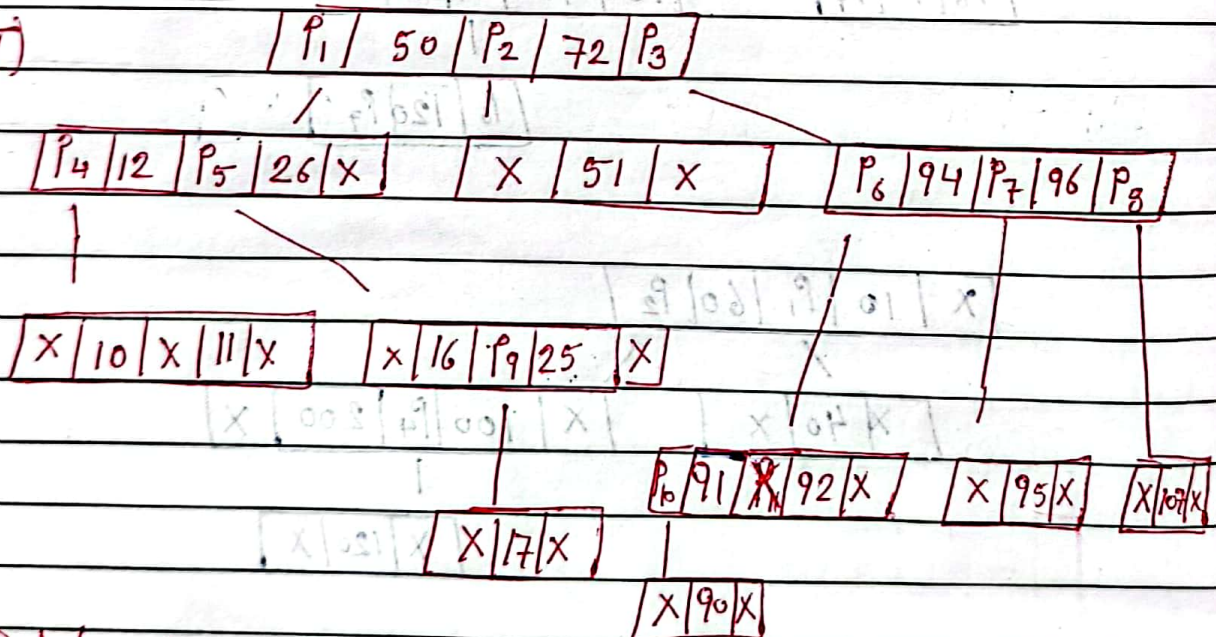
Deletion in M-way tree

I No child

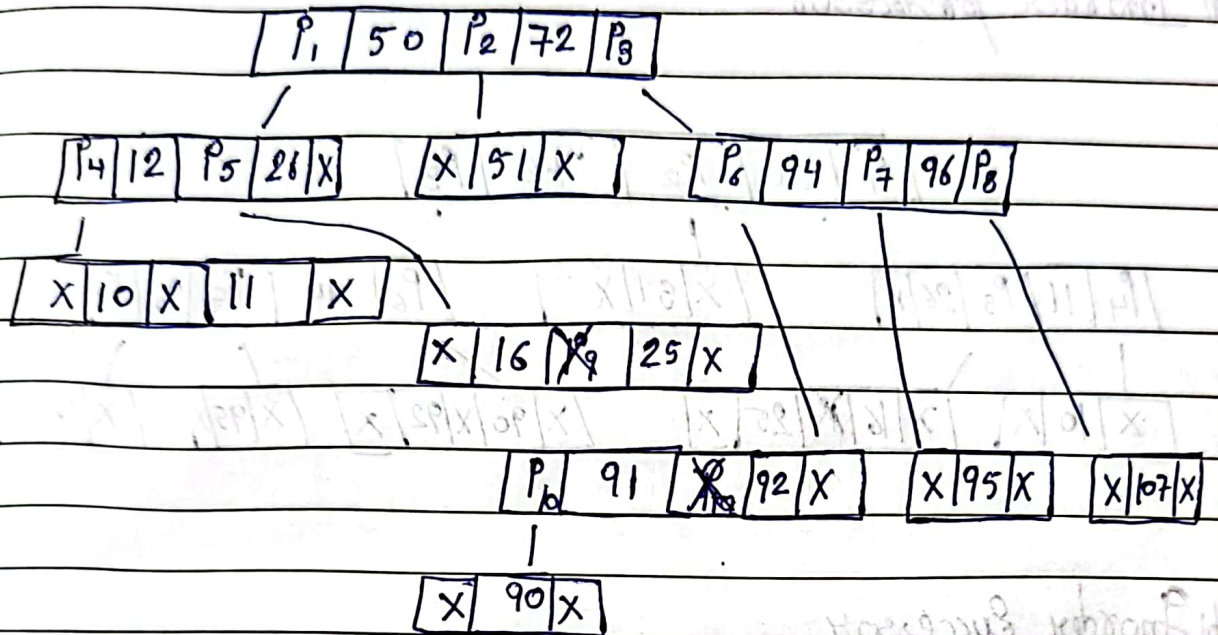
II One child

III Two child

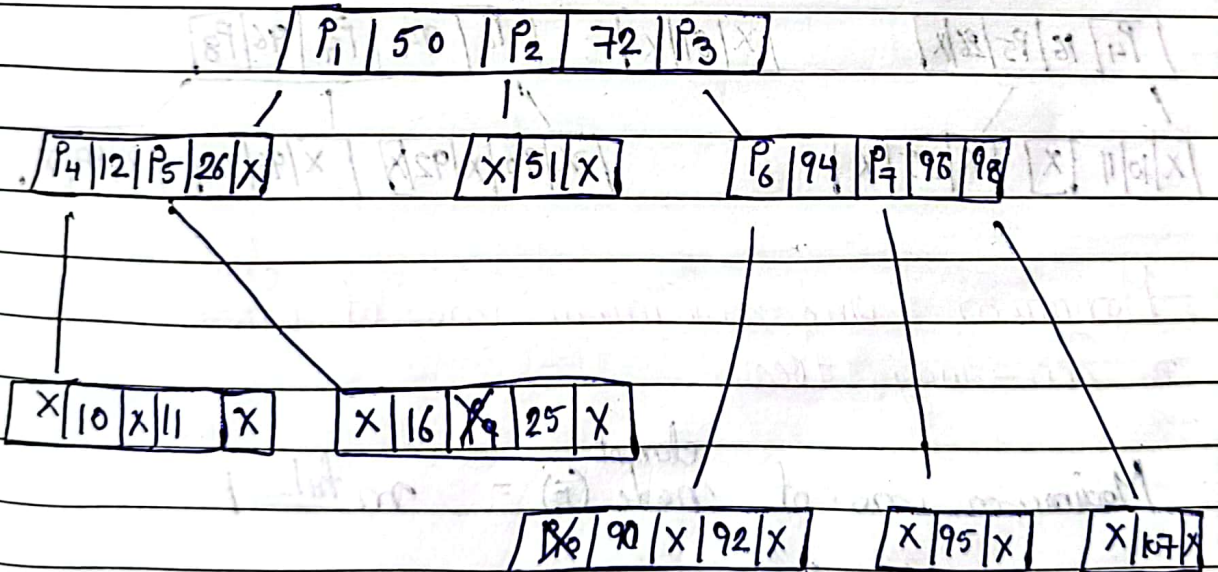
T)



Delete 17



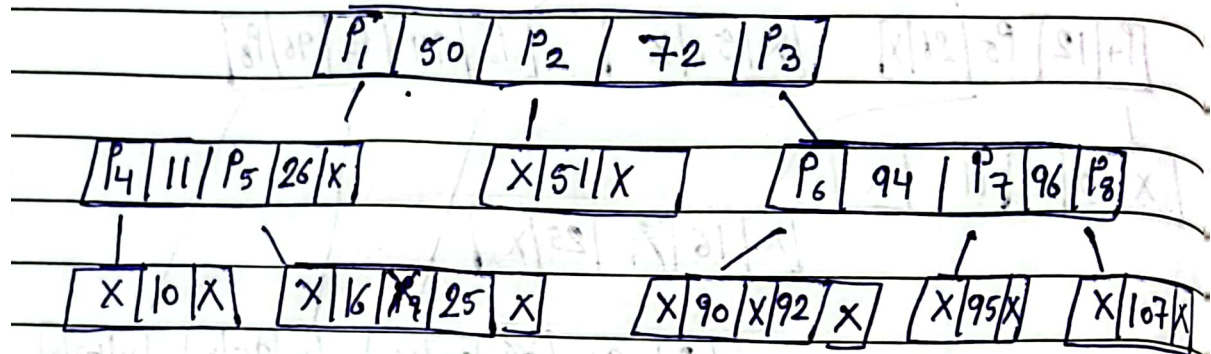
Delete 91



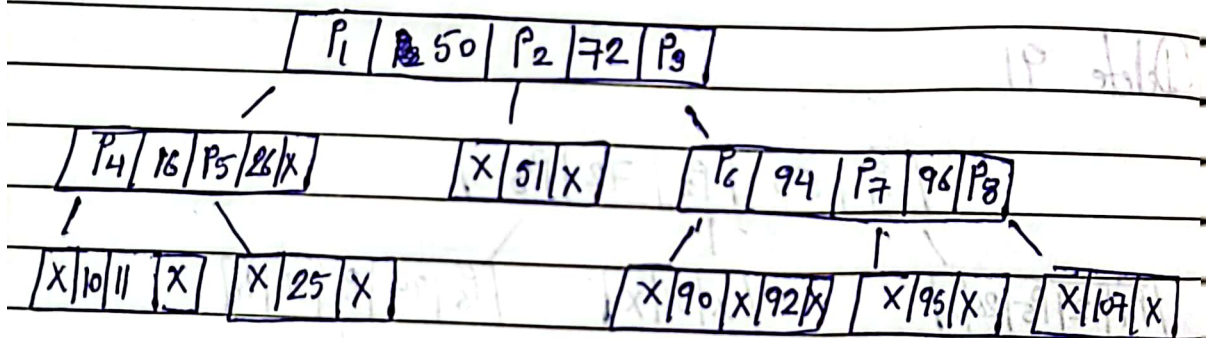
Delete 12

Delete 12

a) Inorder predecessor



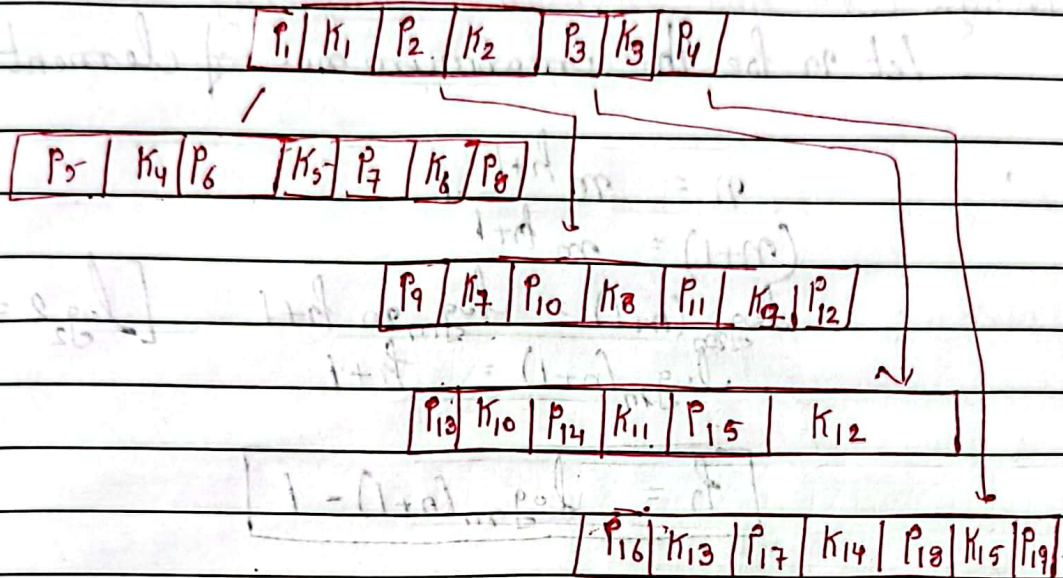
b) Inorder Successor



Maximum and minimum no. of ^{element} ~~node~~ in ~~m~~ m-way tree

Maximum no. of ^{element} ~~node~~ (n) = $m^{h+1} - 1$

Minimum ~~node~~ (n) = $m^{h+1} - 1$
^{element}



$$h=1 \quad m=4$$

$$\text{Total no. of elements} = n(4)^2 - 1$$

$$(\text{Maximum}) = 15$$

$$[P_0 | K_0 | P_1] = [K_2 | P_2]$$

$$[P_3 | K_3 | P_4]$$

$$m=3, \quad h=1$$

$$\text{Total no. of element} = 3$$

$$(\text{Minimum})$$

$$\text{Worst case time complexity} = O(h)$$

$$n = (m-1)h + 1$$

$$= O((m-1)h + 1)$$

$$n-1 = (m-1)h$$

$$= O(mh)$$

$$h = \frac{n-1}{m-1}$$

$$= O(h)$$

$$\frac{(n-1)}{(m-1)}$$

$$= O(n)$$

Average case and Best case complexity

Let n be the maximum no. of element

$$n = m^{h+1} - 1$$

$$(n+1) = m^{h+1}$$

$$\log_m(n+1) = \log_m m^{h+1} \quad [\log_2 2 = 1]$$

$$\log_m(n+1) = h+1$$

$$h = \log_m(n+1) - 1$$

$$h = O(\log_m n)$$

I) Let $m=3$ $n=8$

$$h = \log_3$$

$$\begin{aligned} h &= \log_3(9) - 1 \\ &= \log_3(3^2) - 1 \\ &= 2 - 1 \\ &= 1 \end{aligned}$$

$$\begin{aligned} h &= \log_m n \\ &= \log_3 8 \\ &= 1 \end{aligned}$$

II) Let $m=3$ $n=9$

$$\begin{aligned} h &= \log_3(9+1) - 1 \\ &= \log_3(10) - 1 \\ &= 3 - 1 \\ &= 2 \end{aligned}$$

B- tree

- Min children :- leaf → 0

Root $\rightarrow 2$

internal node $\rightarrow [m_2]$

- Every mode has $\max(m-1)$ keys

Min Keys:- root node - 1

~~all other mode $\rightarrow \left\lfloor \frac{m}{2} \right\rfloor - 1$~~

Q-1 Insert the following keys into B-tree, if order of B-tree is 4.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

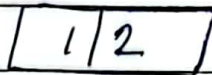
$$\text{Max Keys} = m - 1 = 4 - 1 = 3$$

$$\text{Minimum Keys} = \left\lceil \frac{m}{2} \right\rceil - 1 = \left\lceil \frac{24}{2} \right\rceil - 1 = 11$$

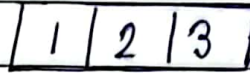
First Key 1



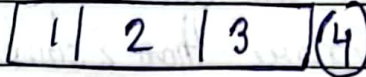
Next Key 2



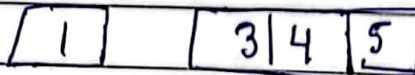
Next Key 3



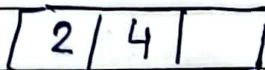
Next Keys 4



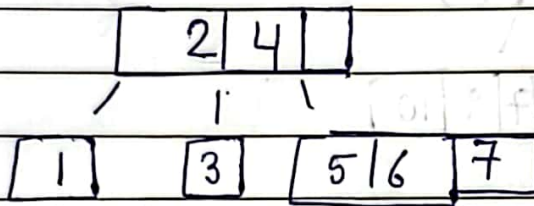
Next Key 5.



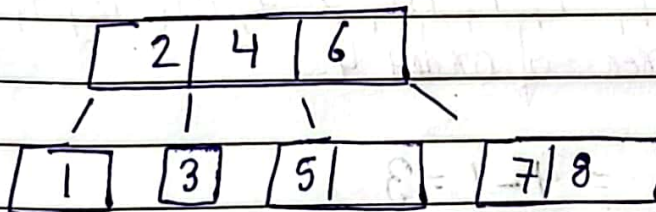
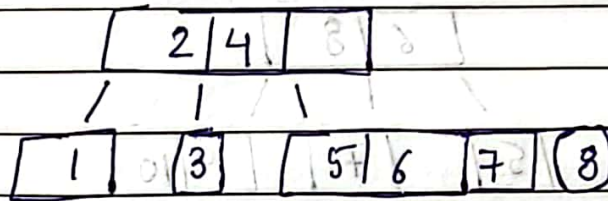
Next Key 6

~~Next Key 7~~

Next Key 7



Next Key 8

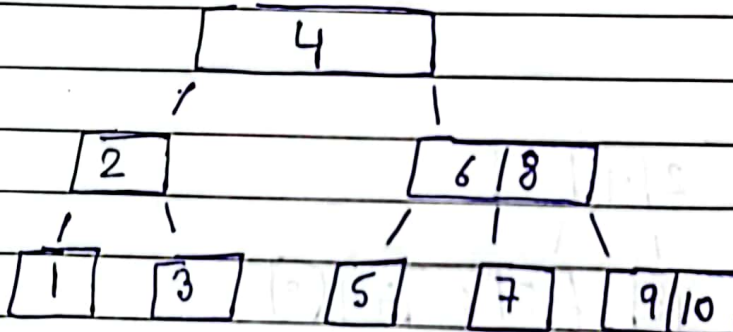
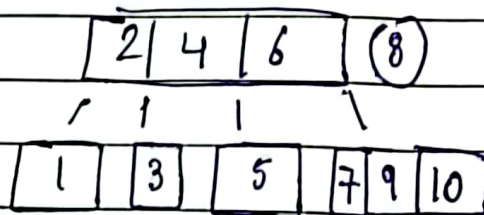


Next Key 9



Next Key 10

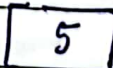




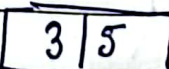
Ex-2 5, 3, 21, 9, 1, 13, 2, 7, 10, 12, 4, 8
Construct a B-tree of order 4

$$\text{Keys} = m-1 = 4-1 = 3$$

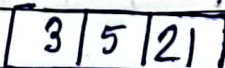
First Key 5



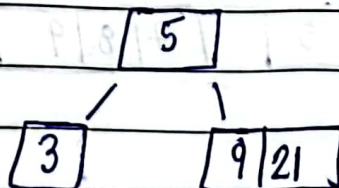
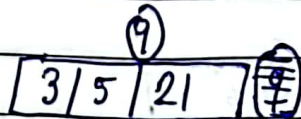
Next Key 3



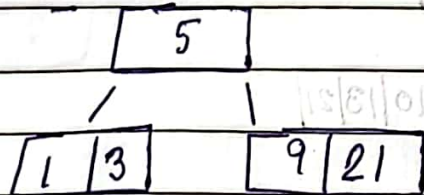
Next Key 21



Next Key 9



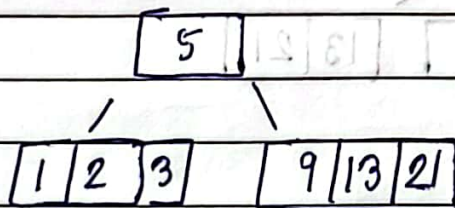
Next Key 1



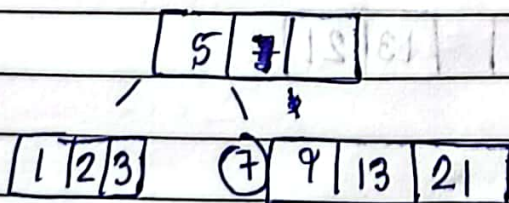
Next Key 13



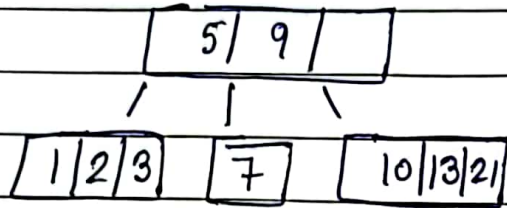
Next Key 2



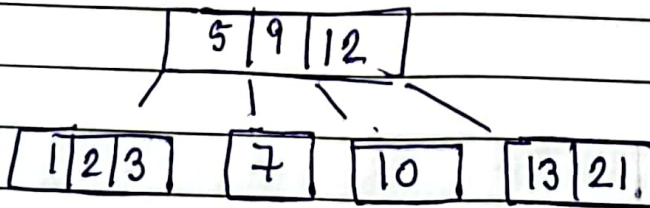
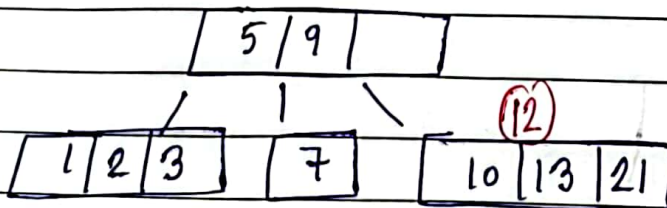
Next Key 7



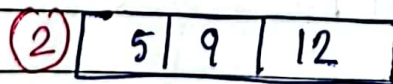
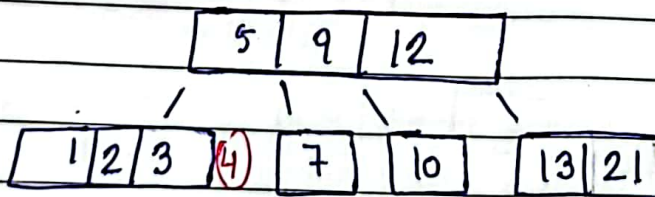
Next element is 10

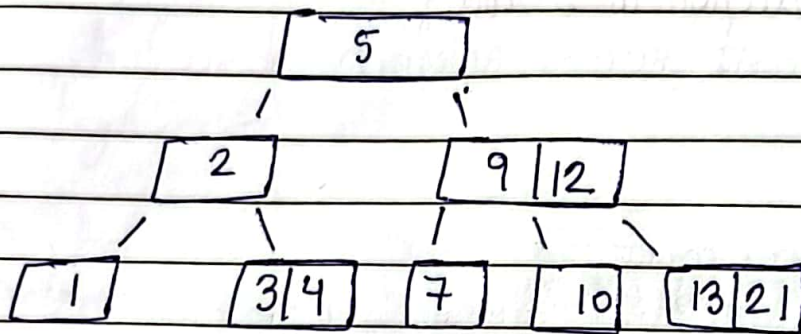


Next Key is 12.



Next Key is 4.



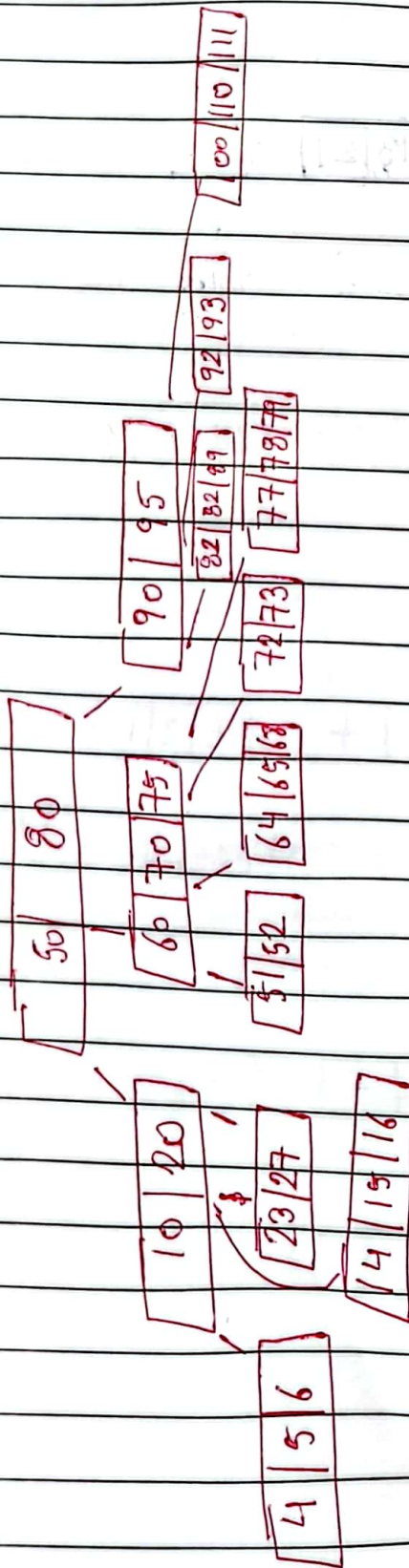


Ex-3 Construct a B-tree of order 5 with the following set of data.

D, H, Z, K, B, P, Q, E, A, S, W, T, C, L, N, Y, M.

$m = 5$

Deletion in B-tree



$$\text{Order}(m) = 5$$

$$\text{Min children} = \left\lceil \frac{m}{2} \right\rceil = \left\lceil \frac{5}{2} \right\rceil = 3$$

$$\text{Max children} = 5$$

$$\text{Min keys} = \left\lceil \frac{m}{2} \right\rceil - 1 = \left\lceil \frac{5}{2} \right\rceil - 1 = 2$$

$$\text{Max keys} = m - 1 = 4$$

Case I) If target Key is in leaf node.

(a) Leaf node contains more than min no. of keys.

(b) Leaf node contains minimum no. of keys.

(1) Borrow from immediate left node (Sibling)

(2) Borrow from immediate right node (Sibling).

(3) Merging the node.

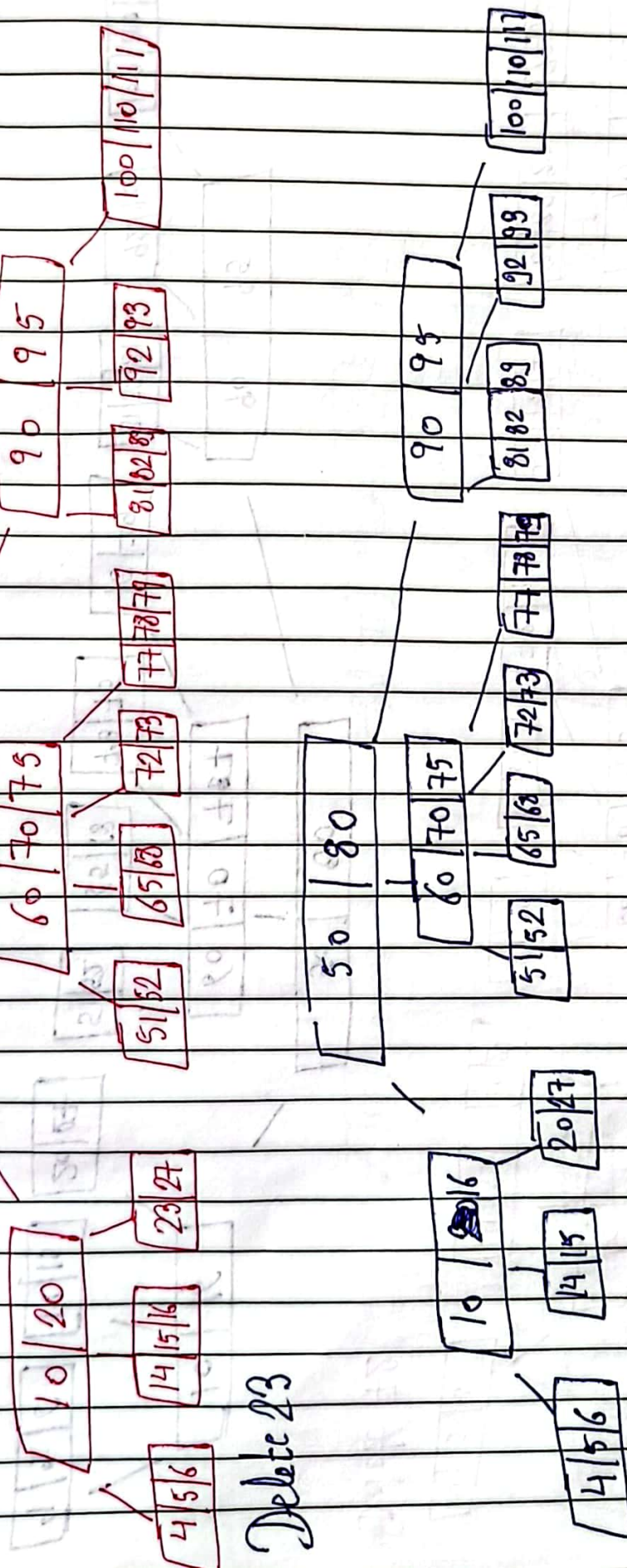
Case II) If target Key is in internal node.

(a) Inorder Predecessor.

(b) Inorder Successor.

(c) Merging the node.

Case I) b) (1)



Delete 23

Case I (B) (2)

4 | 5 | 6

Delete 72

10 | 16

4 | 5 | 6

14 | 15

20 | 27

10 | 16

14 | 15

20 | 27

50 | 80

60 | 70 | 75

51 | 52

65 | 68

72 | 73

77 | 78 | 79

60 | 70 | 75 | 7

51 | 52

65 | 68

73 | 75

78 | 79

81 | 82 | 89

92 | 93

90 | 95

100 | 110 | 111

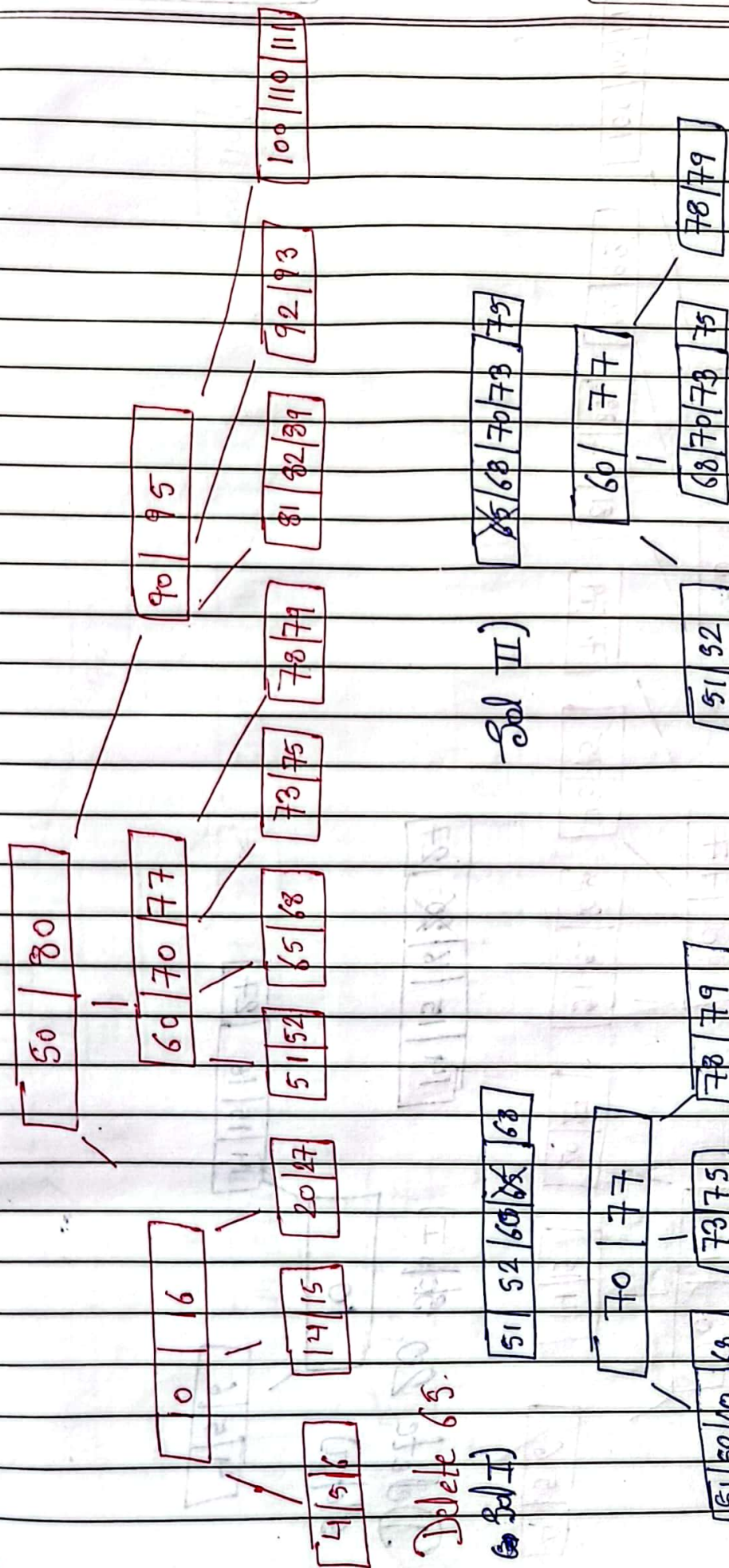
90 | 95

92 | 93

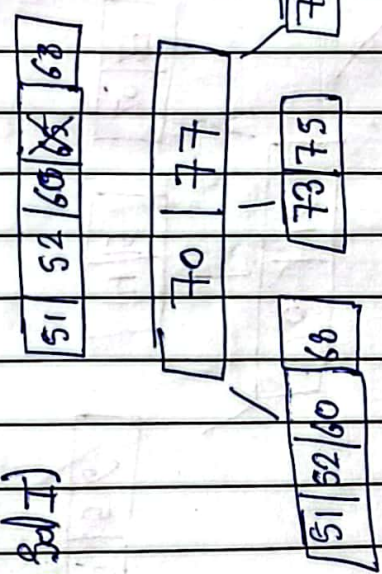
81 | 82 | 89

100 | 110 | 111

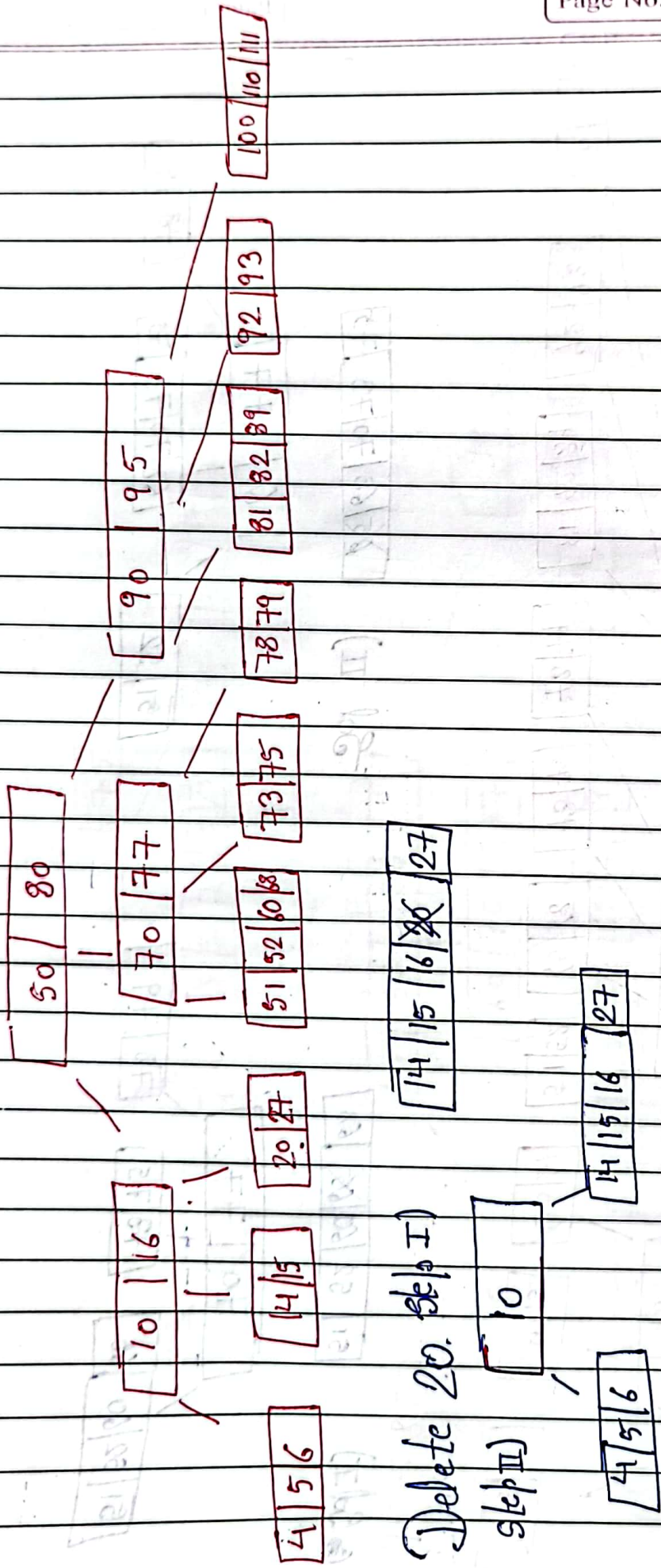
Case I (b) (3)



Sol II)



Case I) b (3)



Step II

10

4 | 5 | 6

14 | 15 | 16 | 17

Step III

10 | 50 | 70 | 77

Step IV

10 | 50 | 70 | 77

14 | 15 | 16 | 17

51 | 52 | 60 | 68

73 | 75

78 | 79

81 | 82 | 89

80

~~50 | 80~~

78 | 79

73 | 75

51 | 52 | 60 | 68

70 | 77

50 | 80

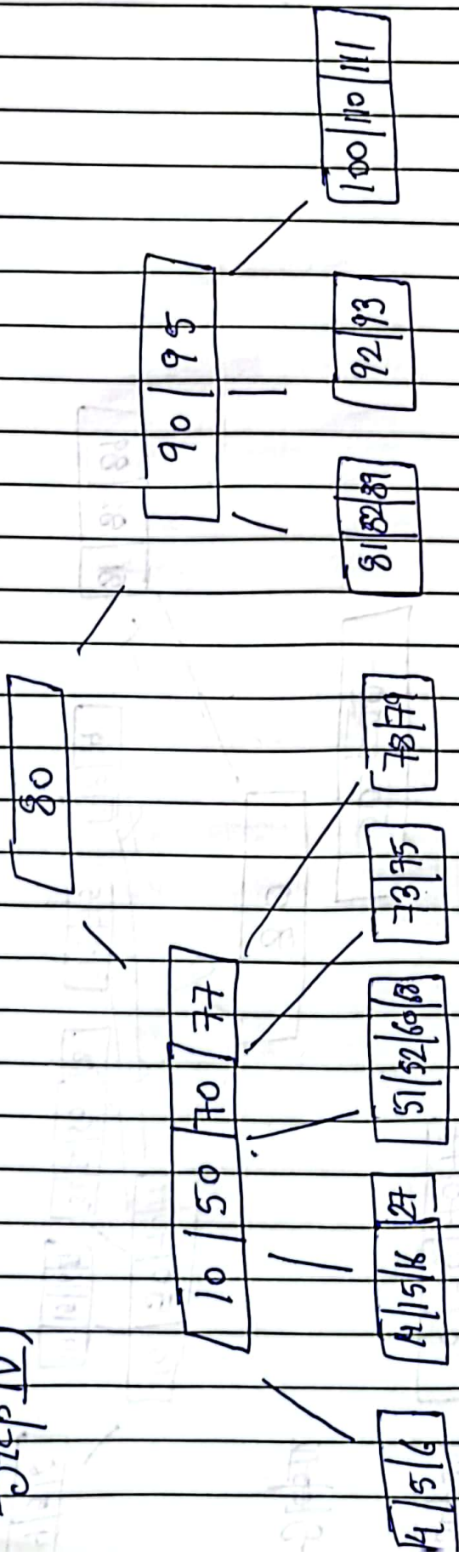
90 | 95

81 | 82 | 89

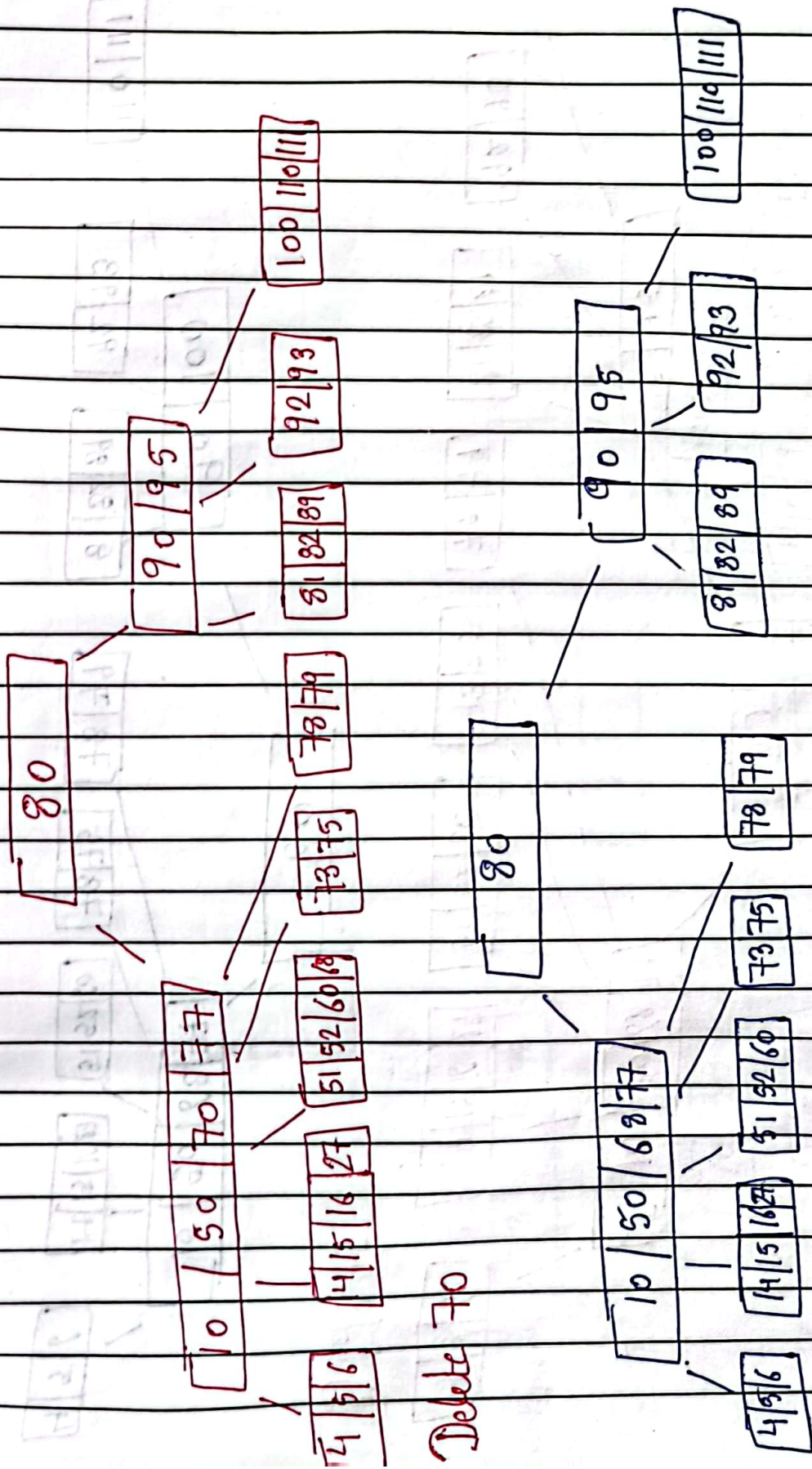
92 | 93

100 | 110 | 111

Step IV)



Case III (a)



Case II (b)

80

10 | 50 | 68 | 77

4 | 5 | 6

14 | 15 | 16 | 17

51 | 52 | 60

73 | 75

78 | 79

81 | 82 | 89

92 | 93

100 | 110 | 111

90 | 95

Delete 95

80

10 | 50 | 68 | 77

4 | 5 | 6

14 | 15 | 16 | 17

51 | 52 | 60

73 | 75

78 | 79

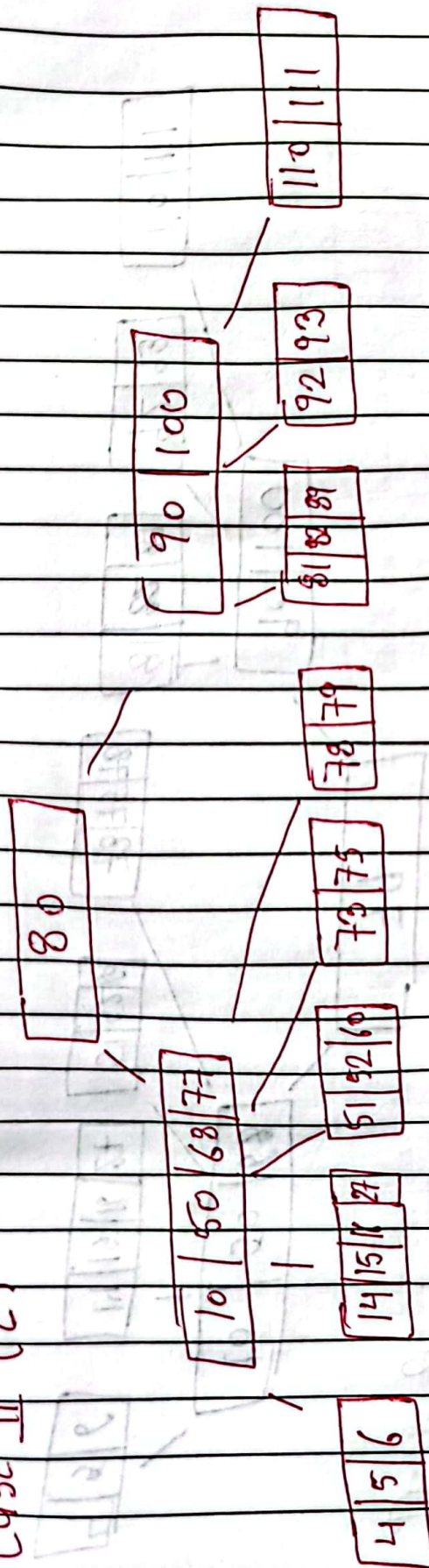
81 | 82 | 89

92 | 93

100 | 110 | 111

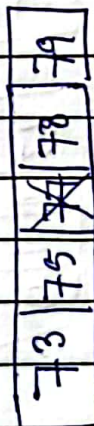
90 | 100

Case II (c)

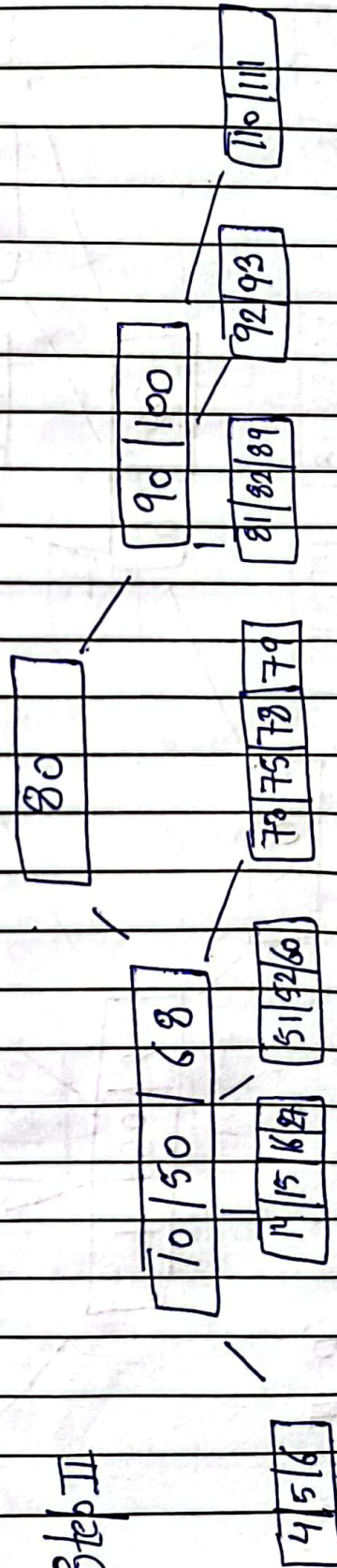


Delete 77

Step I

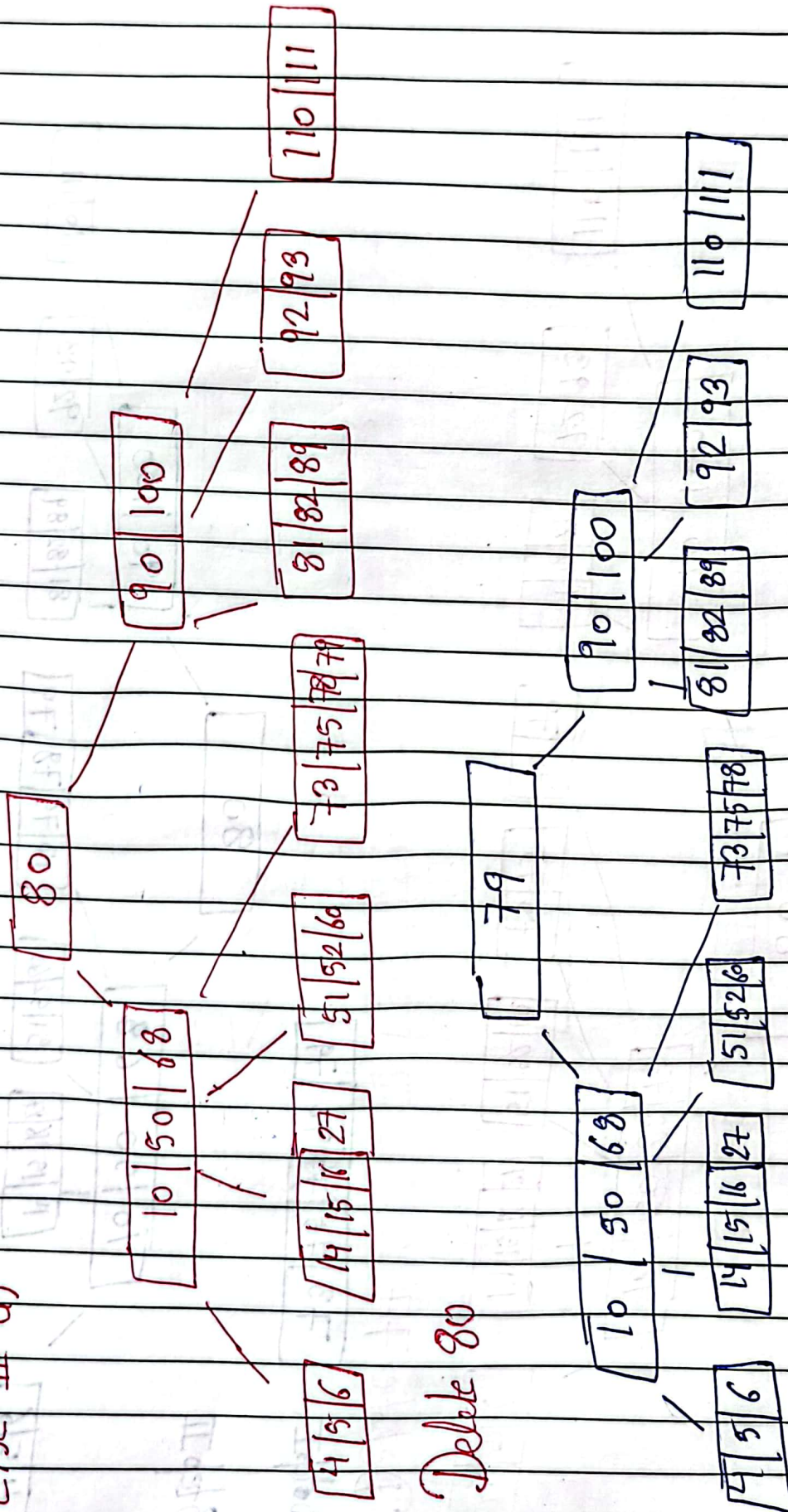


Step II

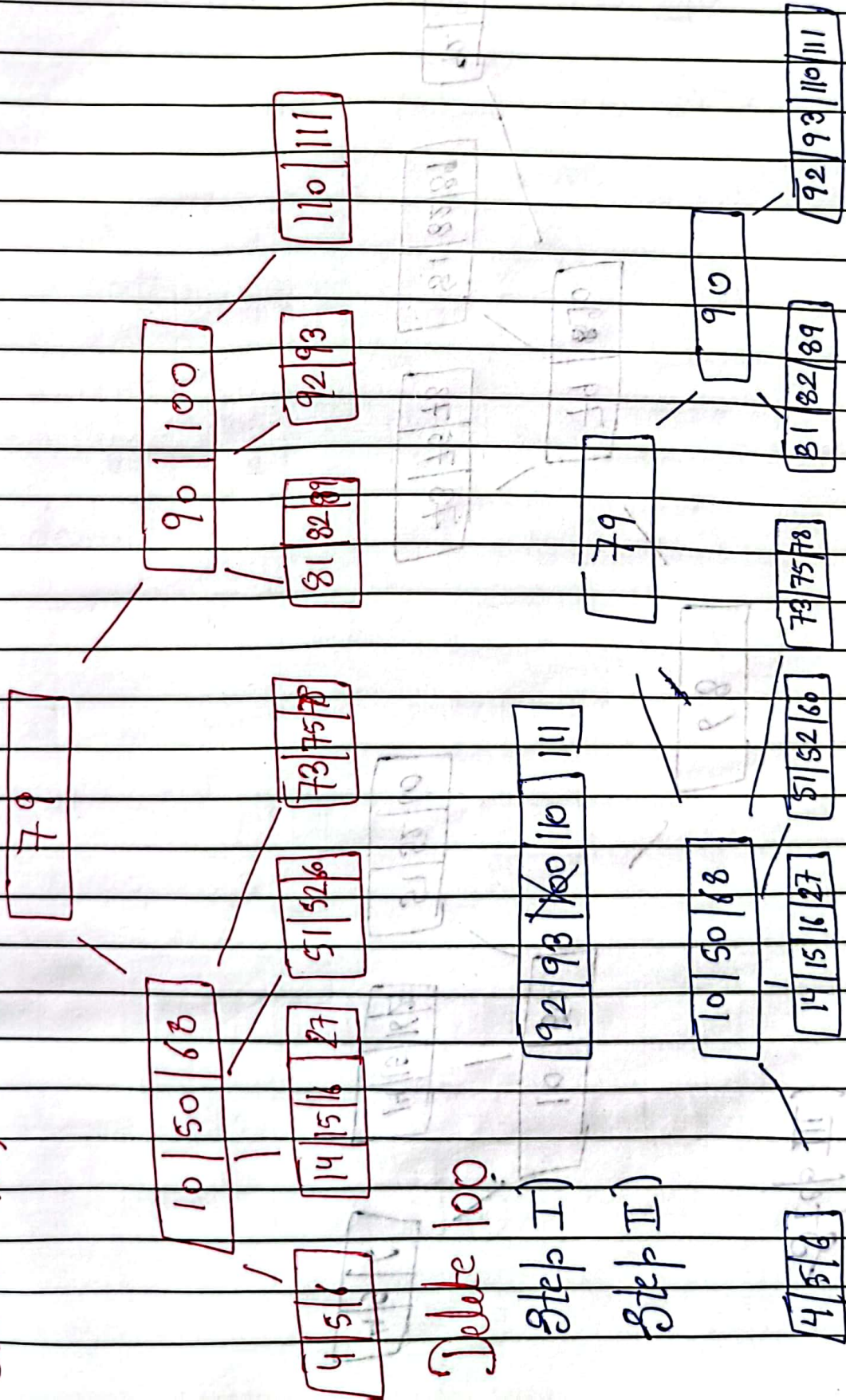


Case II a)

Delete 80



Case II) a) c)



Step III)

68

10 | 50

14 | 15 | 16 | 27

4 | 5 | 6

51 | 52 | 60

79 | 890

73 | 75 | 78

81 | 82 | 89

92 | 93 | 110 | 111

B+ tree

- (1) All the data are stored in leaf node.
 (2) In Internal node pointer of data is stored.

Ex - 1, 4, 7, 10, 17, 21, 31, 25, 19, 20, 28, 42
 Order(m) = 4

$$\text{max children} = 4$$

$$\text{min children} = \lceil \frac{m}{2} \rceil = \lceil \frac{4}{2} \rceil = 2$$

$$\text{max Keys} = (m-1) = 3$$

$$\text{min Keys} = \lceil \frac{m}{2} \rceil - 1 = \lceil \frac{4}{2} \rceil - 1 = 1$$

First element is 1

1

Next element is 4

1 | 4

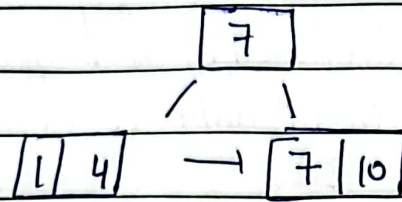
Next element is 7

1 | 4 | 7

Next element is 10

1 | 4 | 7 | 10

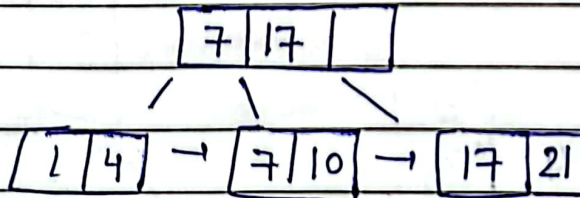
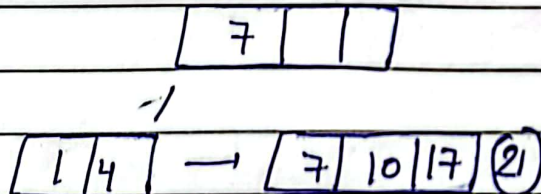
1, 4, 7, 10, 17, 21, 31, 25, 19, 20, 28, 42



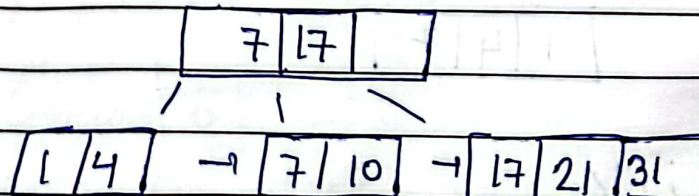
Next element is 17



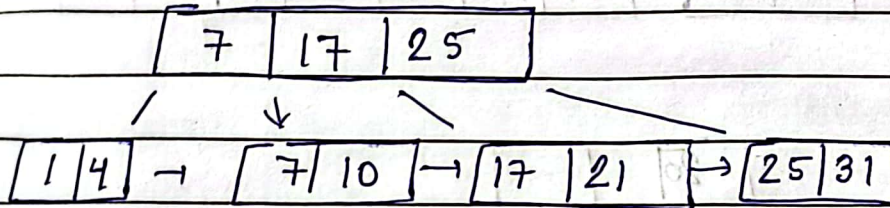
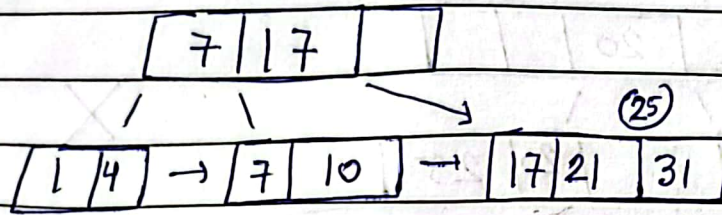
Next element 21



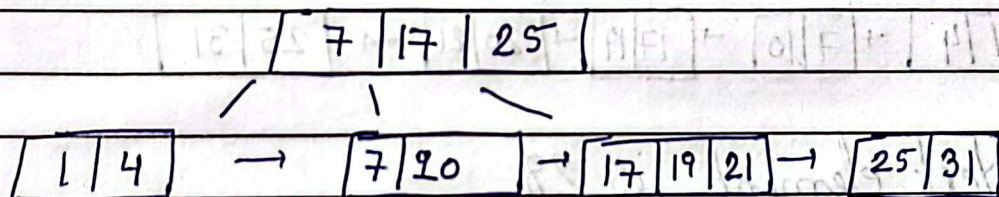
Next element is 31



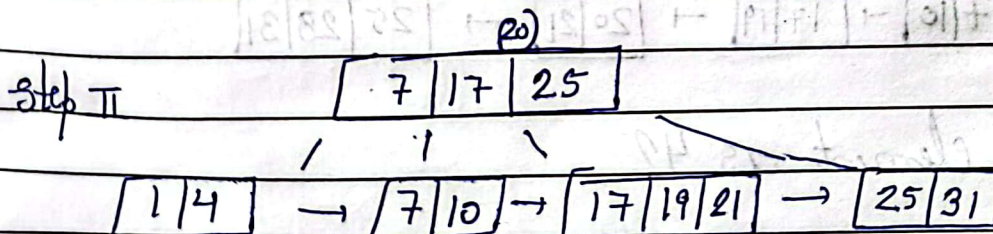
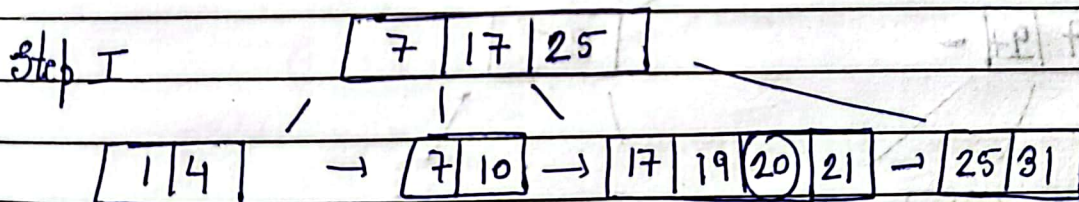
1, 4, 7, 10, 17, 21, 25, 19, 20, 28, 42
 Next element is 25



Next element is 19

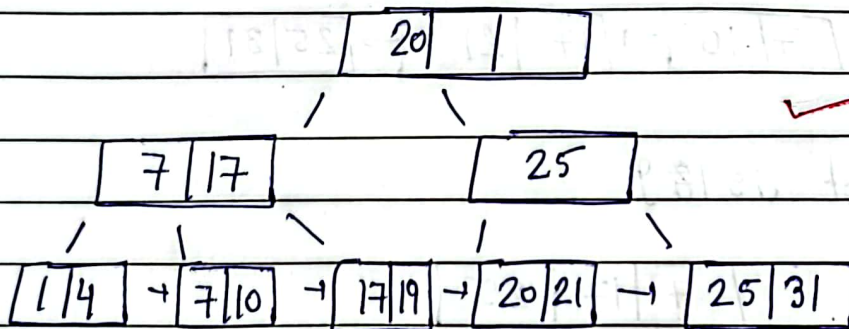
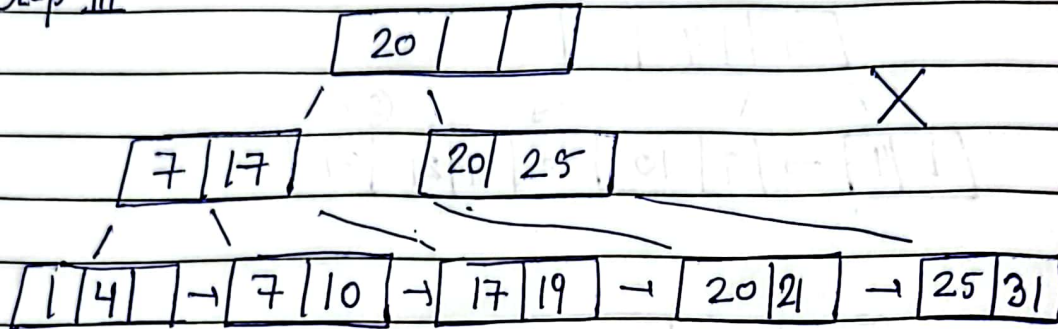


Next element is 20

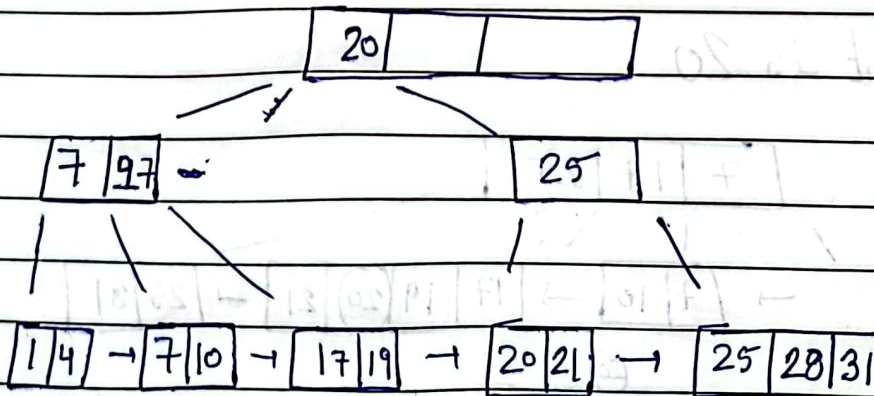


1, 4, 7, 10, 17, 21, 31, 25, 19, 20, 28, 42

Step III



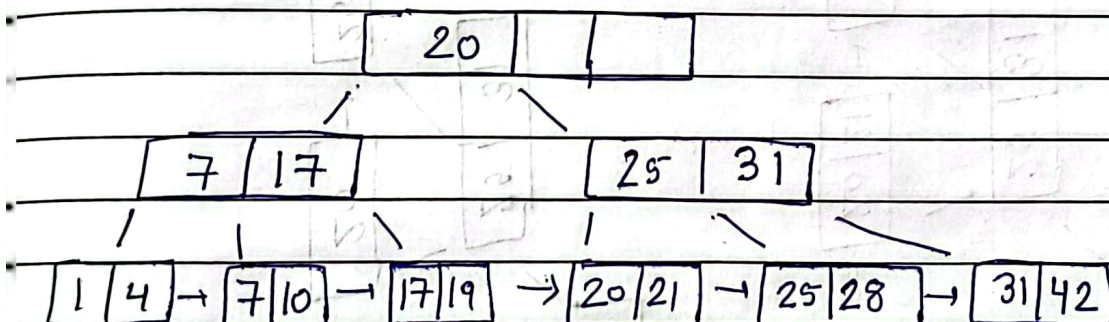
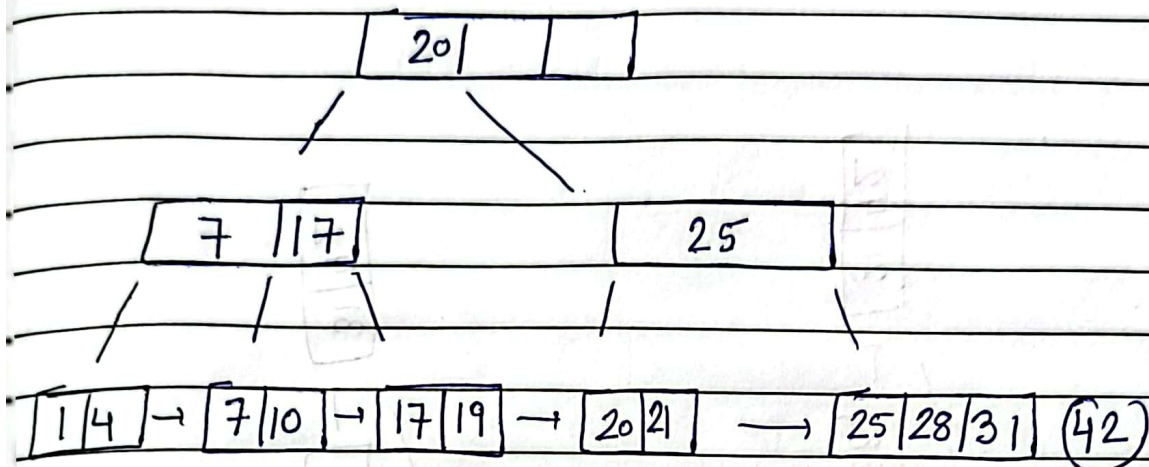
Next element is 28



Next element is 42



1, 4, 7, 10, 17, 21, 31, 25, 19, 20, 28, 42
 Next element is 42



Deletion in B+ tree

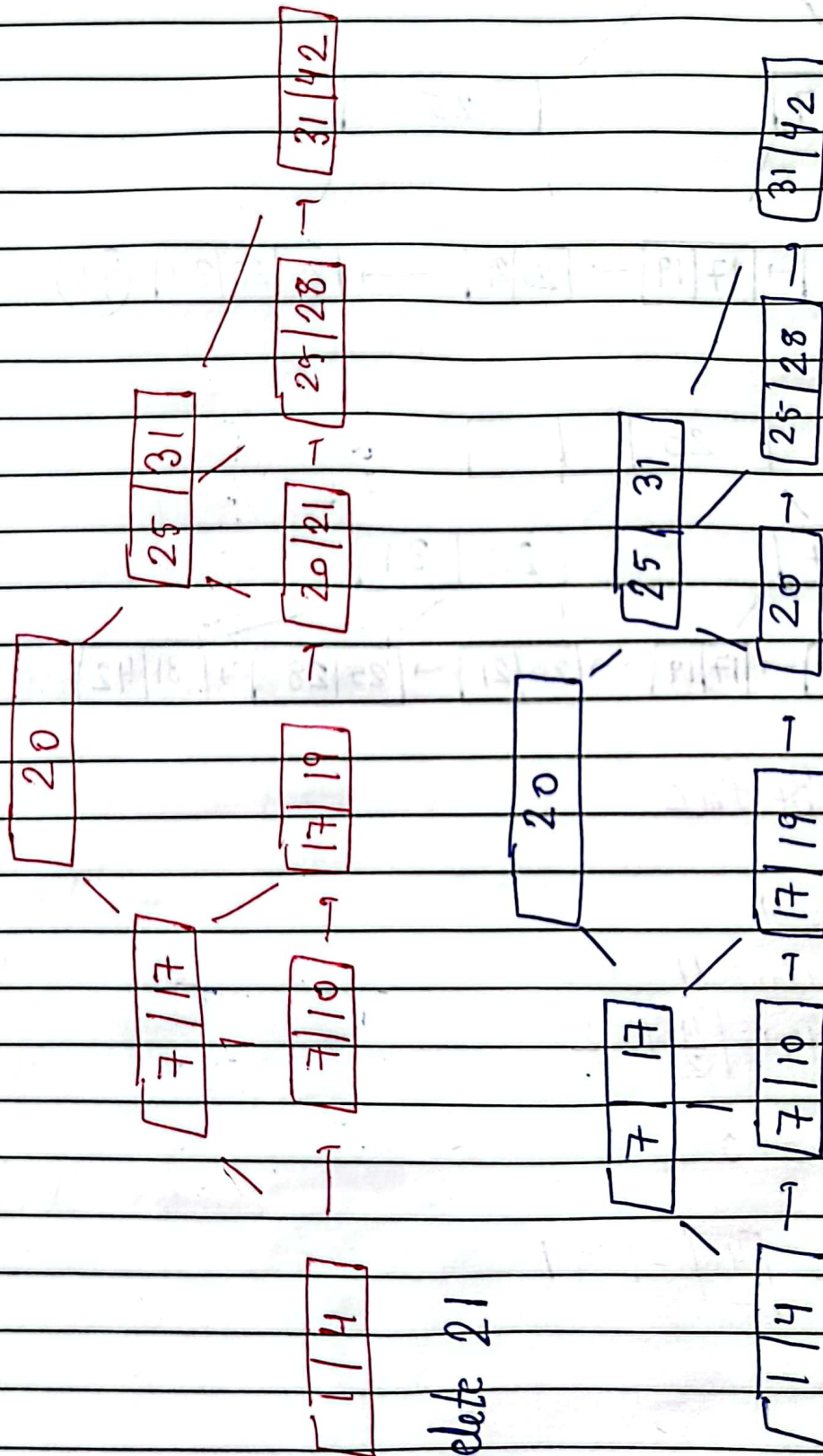
$$\text{Order}(m) = 4$$

$$\text{Max children} = 4$$

$$\text{Min children} = \lceil \frac{4}{2} \rceil = 2$$

$$\text{Max Keys} = 3$$

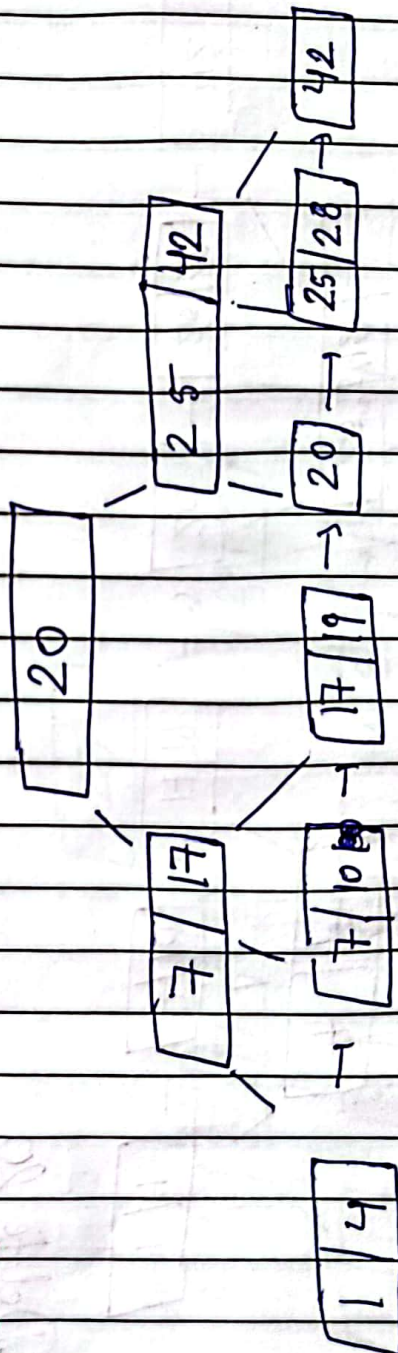
$$\text{Min Keys} = \lceil \frac{4}{2} \rceil - 1 = 1$$

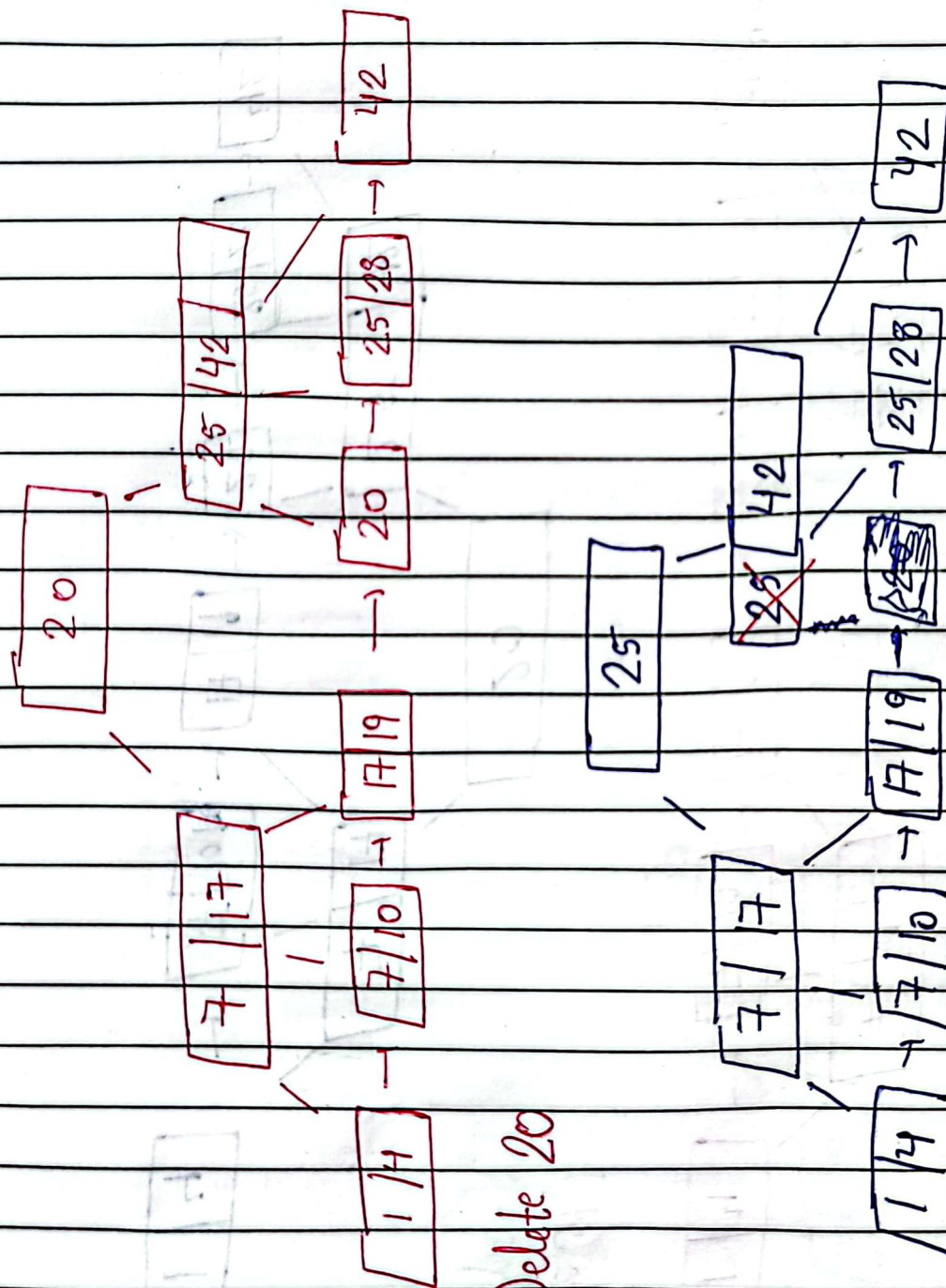


Delete 21

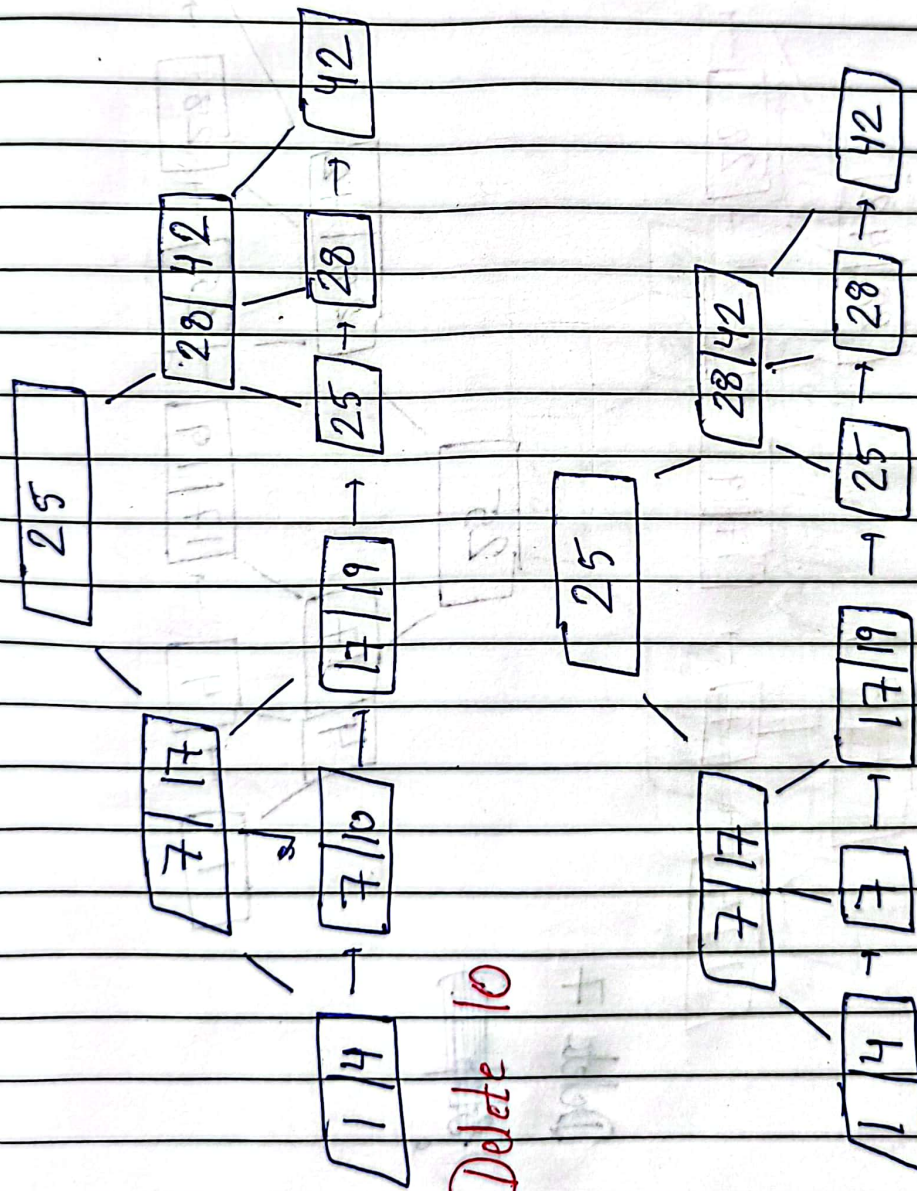


Delete 31

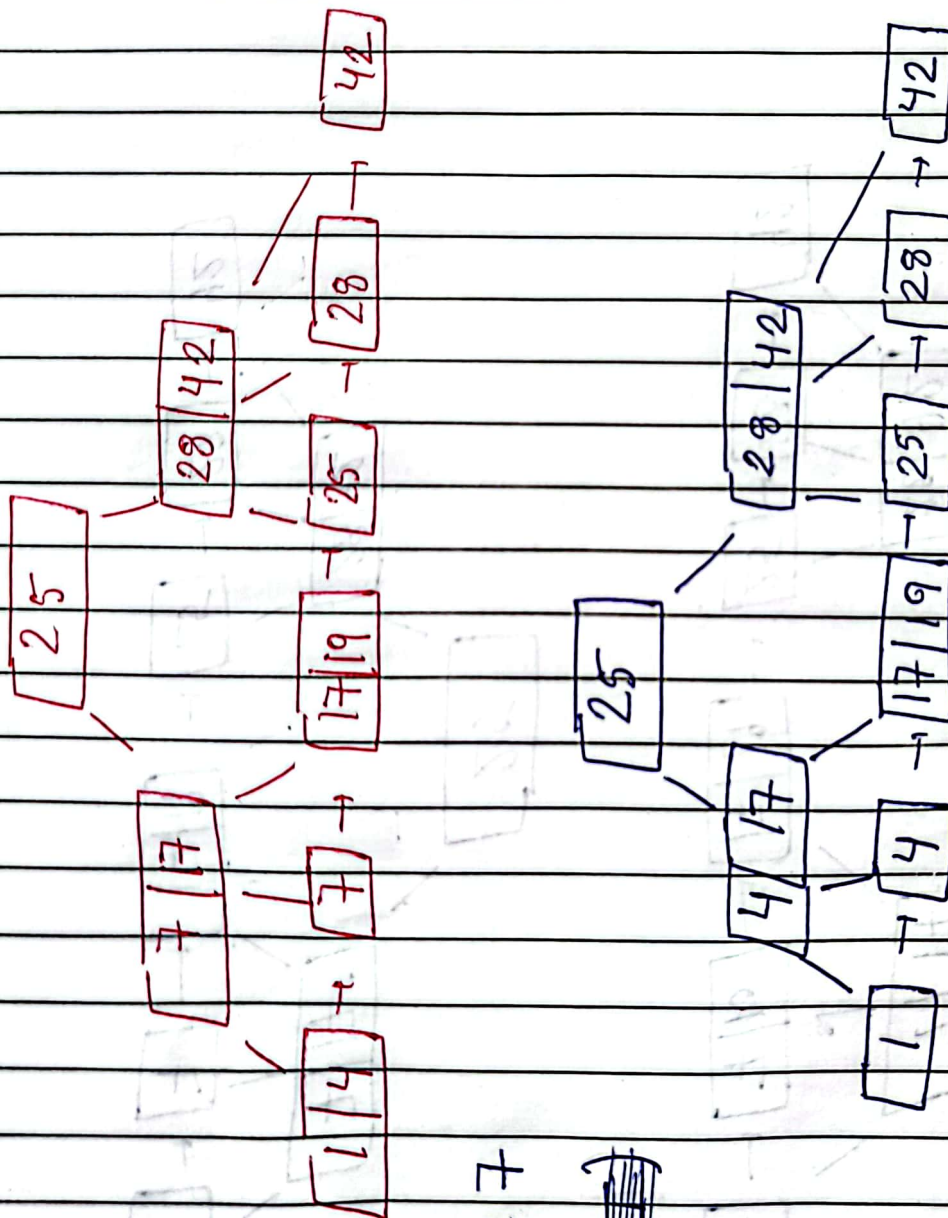




Delete 20



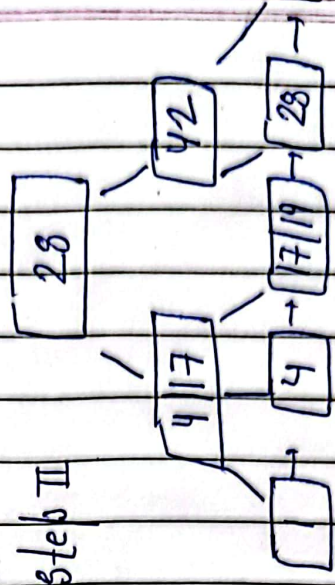
Delete 10



Delete 7

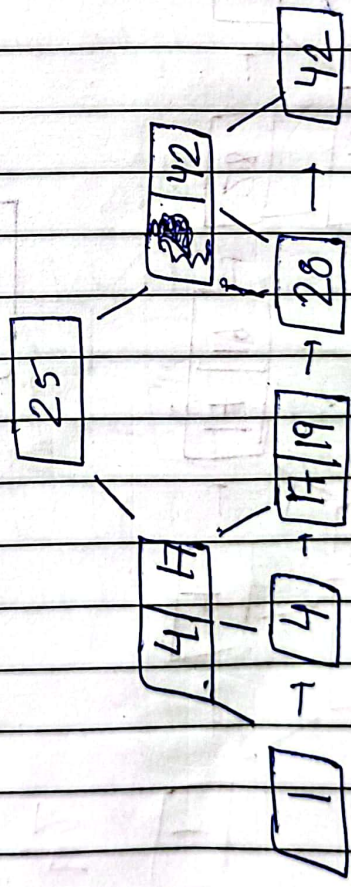
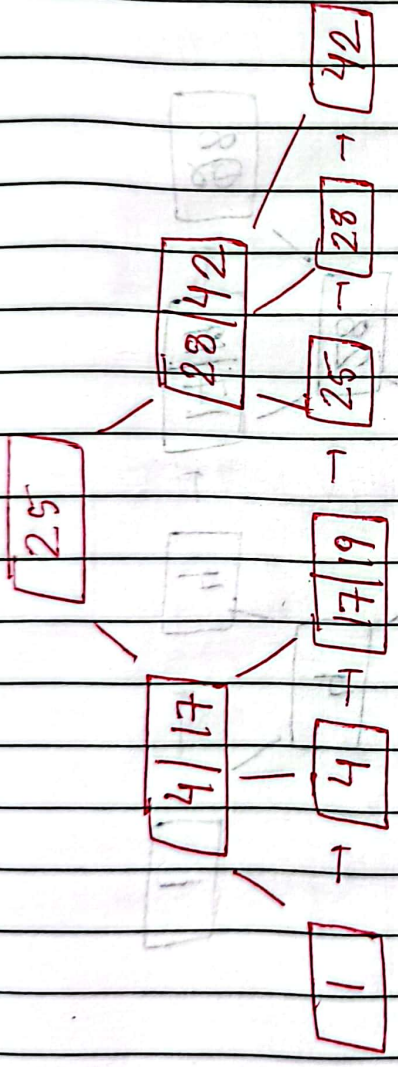


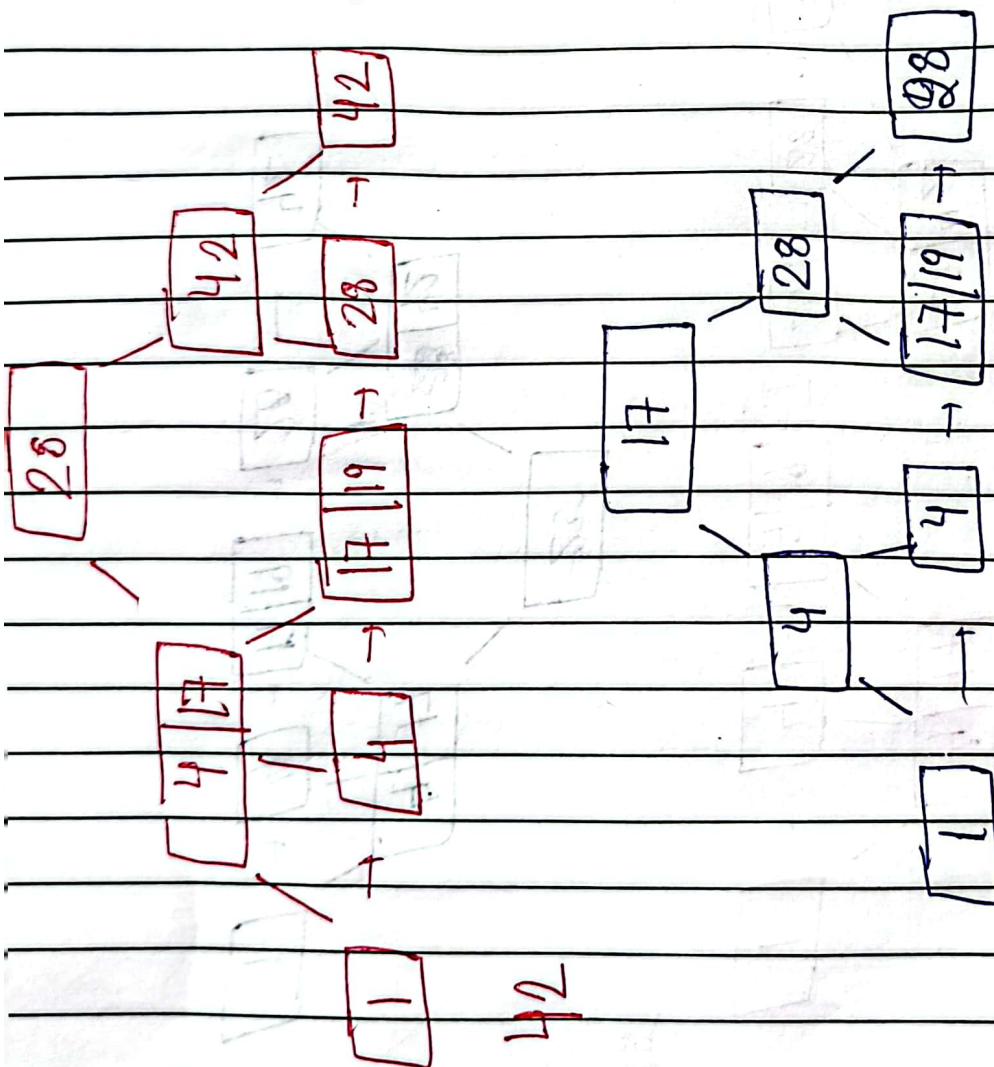
Step II



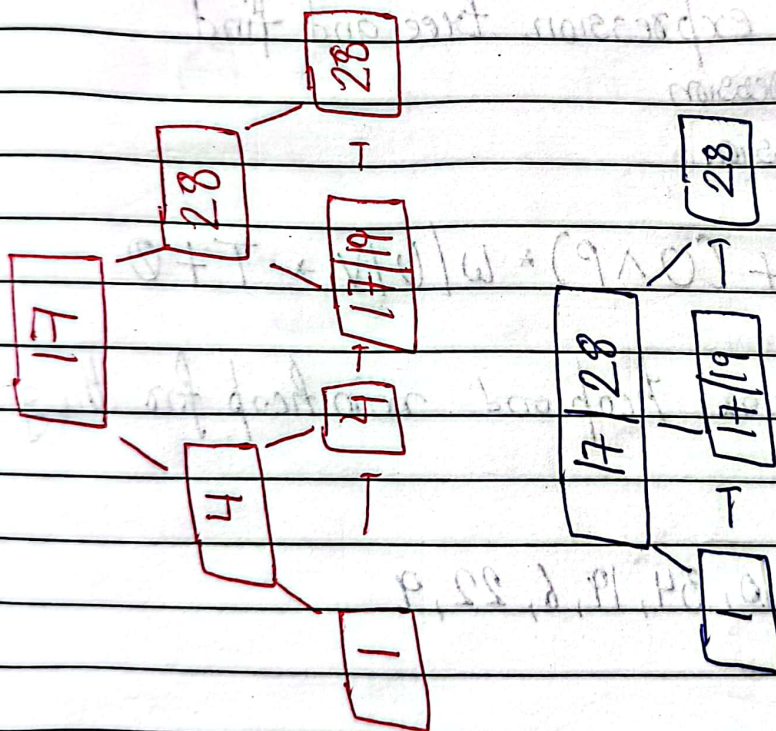
Delete 25

Step I)





Delete 42



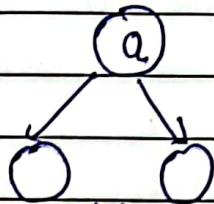
Delete 4

2-3 tree

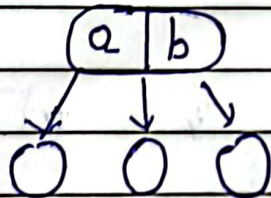
It contains 3 different kinds of nodes.

- a) leaf node b) 2-node c) 3-node

2-node $\rightarrow$ 2-children and one data element,



3-node $\rightarrow$ 3-children and 2 data elements

Properties

- 1) Balanced tree
- 2) Data is stored in sorted manner
- 3) Every 2 node or 3 node
- 4) All leaf node must be at same level

Construct 2-3 tree

Ex - 50, 30, 10, 70, 60

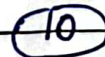
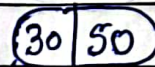
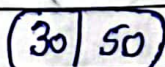
Insert 50



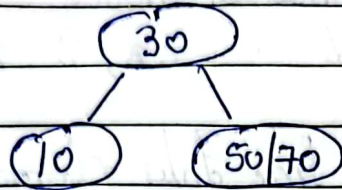
Insert 30



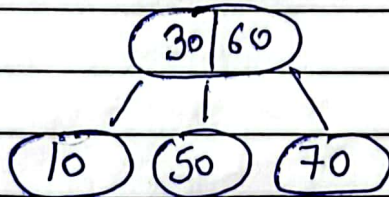
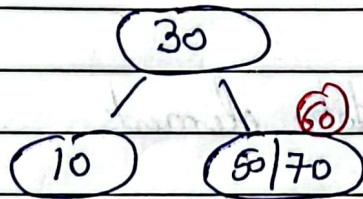
Insert 10



Insert 70



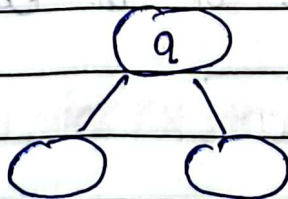
Insert 60



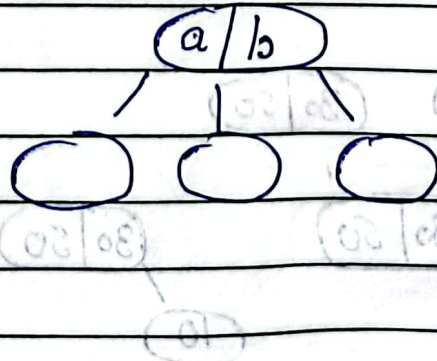
2-3-4 tree.

- Each node contain 2-node, 3-node, 4-node.

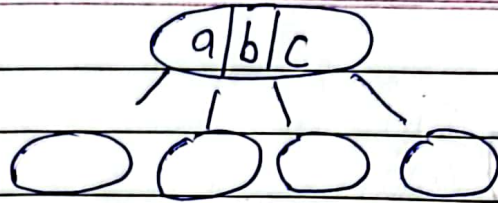
2-node



3-node



4-mode



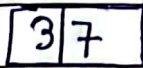
Every leaf node will be at some level.
It is used to make dictionaries.

Example 3, 7, 4, 9, 10, 0, 5, 6, 8, 2, 1

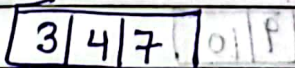
Insert 3



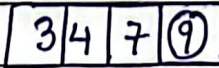
Next key 7



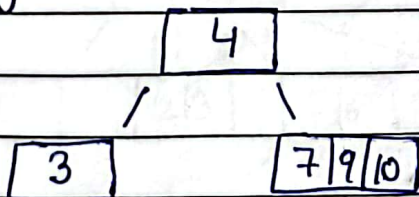
Next key 4



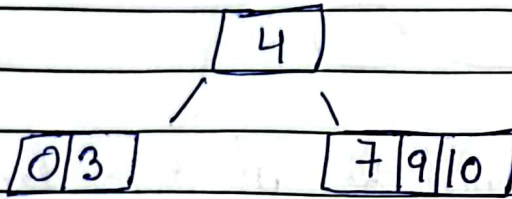
Next key 9



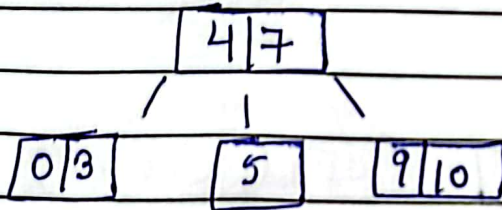
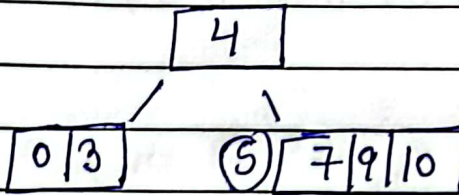
Next key 10



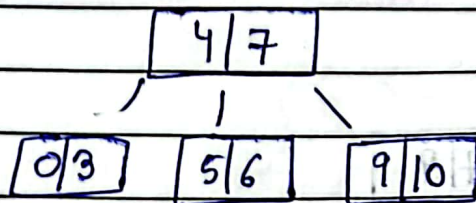
Next Key is 0



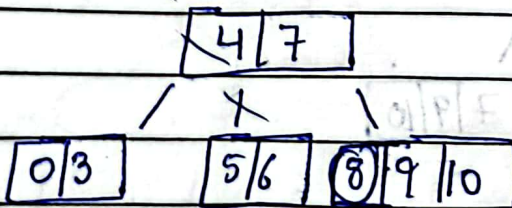
Next Key is 5



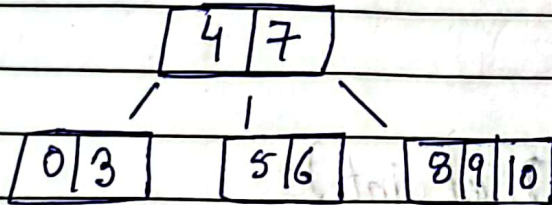
Next Key is 6



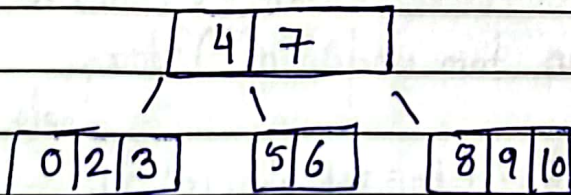
Next Key is 8



Next Key is 8



Next element is 2



Next element is 1

